2: Coherent perfect absorption by a single nanoparticle with an azimuthally symmetric incident beam

This set of notes is on the paper Perfect absorption of a focused light beam by a single nanoparticle (https://onlinelibrary.wiley.com/doi/abs/10.1002/lpor.202000430?casa_token=Wc0i7Q2Kf9EAAAAA:eyrI3Zs-eBJDO36d_QvW7vMngLaB3btUd770cWKkcJDdmKKg9fYZM8bLoQhAQDklaAB8OQwj7efc-Tw) (**Laser and Photonics Reviews, 2021**). This paper is closely related to a conference paper for which we already wrote up notes here: https://hackmd.io/@aligho/H1iiWa2Jfx (https://hackmd.io/@aligho/H1iiWa2Jfx).

Section 1: The incident beam

$$\mathbf{E}_{k_\rho}^{inc}(\rho,z) = \frac{E_2(k_\rho)}{k_\rho}\nabla\times\Big[\mathbf{e}_z J_0(k_\rho\rho)e^{-ik_z z}\Big] + \frac{E_1(k_\rho)}{k_\rho^2}\nabla\times\nabla\times\Big[\mathbf{e}_z J_0(k_\rho\rho)e^{-ik_z z}\Big]$$

$$= \tilde{E}_2(k_\rho,z)\begin{bmatrix}0\\-J_0'(k_\rho\rho)\\0\end{bmatrix} + \tilde{E}_1(k_\rho,z,\theta)\begin{bmatrix}-i(k_z/k_\rho)J_0'(k_\rho\rho)\\0\\(1/k_\rho\rho)\Big[-\partial_\rho(\rho J_0'(k_\rho\rho))\Big]\end{bmatrix},$$

where we have defined $\tilde{E}_{1,2}(k_\rho,z)\equiv E_{1,2}(k_\rho)e^{-ik_z z}$. We also note that $\frac{1}{x}\partial_x(xJ_0'(x)) = J_0''(x)+\frac{1}{x}J_0'(x) = -J_0'(x)$. Thus, we may write the incident beam as follows:

$$\mathbf{E}_{k_\rho}^{inc}(\rho,z) = \tilde{E}_2(k_\rho,z)\begin{bmatrix}0\\-J_0'(k_\rho\rho)\\0\end{bmatrix} + \tilde{E}_1(k_\rho,z,\theta)\begin{bmatrix}-i(k_z/k_\rho)J_0'(k_\rho\rho)\\0\\J_0(k_\rho\rho)\end{bmatrix},$$

Note that at $\rho=0$, the first term above goes to zero. Thus, we need only keep the second term to find the field that engenders the dipole moment of the scatterer.

Section 2: The reflected beam

For a PEC, we require the the tangential component of the electric field to vanish at the PEC interface. Thus, we uniquely obtain the reflected beam to be given by:

$$\mathbf{E}^{\text{ref}}(\rho, z) = E_1(k_\rho)\begin{bmatrix} i(k_z/k_\rho)J_0'(k_\rho \rho) \\ 0 \\ J_0(k_\rho \rho)\end{bmatrix} e^{ik_z z} e^{2ik_z h},$$

which is exactly **Equation S2** of the Supplement of the paper.

Section 3: The scattered field

We use the Dyadic Green's function:

$$\mathbf{E}_{\text{dipole}}(\mathbf{r}) = \frac{1}{\varepsilon_0}\left[\frac{\omega^2}{c^2} + \nabla\nabla\right]\mathrm{G}(\mathbf{r}, \mathbf{r}')\mathbf{p}(\mathbf{r}')$$

Together with the Sommerfeld expansion:

$$\frac{e^{ikr}}{4\pi r} = \frac{i}{4\pi}\int_0^\infty \frac{J_0(k_\rho \rho)e^{i|z|\sqrt{k_0^2 - k_\rho^2}}}{\sqrt{k_0^2 - k_\rho^2}} k_\rho \mathrm{d}k_\rho,$$

Thus, the scattered field is given by:

$$\frac{p}{4\pi\varepsilon_0}\int_0^\infty \left[\frac{ik_0^2}{k_z}J_0(k_\rho \rho)e^{ik_z|z|}\mathbf{e}_z \mp \nabla J_0(k_\rho \rho)e^{ik_z|z|}\right]k_\rho \mathrm{d}k_\rho =$$
$$\frac{p}{4\pi\varepsilon_0}\int_0^\infty \left[\frac{ik_\rho^2}{k_z}J_0(k_\rho \rho)\mathbf{e}_z \mp k_\rho J_0'(k_\rho \rho)\mathbf{e}_\rho\right]e^{ik_z|z|}k_\rho \mathrm{d}k_\rho,$$

which is equivalent to **Equation S4** of the Supplement of the paper.

Section 4: The reflected field associated with the scattered field

By evaluating the scattered field at $z = -h$, we immediately see that the necessary reflected field is given by:

$$\frac{p}{4\pi\varepsilon_0}\int_0^\infty \left[\frac{ik_\rho^2}{k_z}J_0(k_\rho \rho)\mathbf{e}_z - k_\rho J_0'(k_\rho \rho)\mathbf{e}_\rho\right]e^{ik_z z}e^{2ik_z h}k_\rho \mathrm{d}k_\rho,$$

which is **Equation S5** of the Supplement of the paper.

Section 5: The effective polarizability

$$\mathbf{p} = p\mathbf{e}_z = \varepsilon_0\alpha_0\left[\frac{p}{4\pi\varepsilon_0}\int_0^\infty \frac{ik_\rho^3}{k_z}e^{2ik_\rho h}\mathrm{d}k_\rho\mathbf{e}_z + \mathbf{E}^{\mathrm{inc}} + \mathbf{E}^{\mathrm{ref}}\right] \rightarrow$$

$$p\mathbf{e}_z\left[1 - \frac{i\alpha_0}{4\pi}\int_0^\infty \frac{k_\rho^3}{k_z}e^{2ik_zh}\mathrm{d}k_\rho\right] = \varepsilon_0\alpha_0\left[\mathbf{E}^{\mathrm{inc}} + \mathbf{E}^{\mathrm{ref}}\right]$$

From this, we obtain the effective polarizability:

$$\alpha^{-1} = \left[\frac{1}{\alpha_0} - \frac{i}{4\pi}\int_0^\infty \frac{k_\rho^3}{k_z}e^{2ik_zh}\mathrm{d}k_\rho\right]^{-1},$$

which is consistent with **Equation S10** of the Supplement of the paper.

Section 6: The CPA condition

We require the sum of the reflected, scattered and reflected scattered fields to be zero. This gives us the following:

$$E_1(k_\rho)\begin{bmatrix} i(k_z/k_\rho)J_0'(k_\rho\rho) \\ 0 \\ J_0(k_\rho\rho)\end{bmatrix}e^{2ik_zh} + \frac{p(1+e^{2ik_zh})}{4\pi\varepsilon_0}\begin{bmatrix} -k_\rho^2 J_0'(k_\rho\rho) \\ 0 \\ (ik_\rho^3)J_0(k_\rho\rho)/(k_z)\end{bmatrix} = 0$$

This gives us the following condition:

$$E_1(k_\rho)e^{2ik_zh} + \frac{ik_\rho^3}{k_z}\frac{p}{4\pi\varepsilon_0}(1+e^{2ik_zh}) = 0,$$

which is **Equation S12** of the Supplement of the paper.

We now just need to find p in terms of the incident field:

$$\mathbf{p} = p\mathbf{e}_z = \varepsilon_0\alpha_0\left[\frac{p}{4\pi\varepsilon_0}\int_0^\infty \frac{ik_\rho^3}{k_z}e^{2ik_\rho h}\mathrm{d}k_\rho\mathbf{e}_z + \int_0^{k_0}\left[1+e^{2ik_zh}\right]E_1(k_\rho)\mathrm{d}k_\rho\mathbf{e}_z\right]$$

Equivalently, we may write this in terms of the effective polarizability:

$$p = \varepsilon_0\alpha\int_0^{k_0}\left[1+e^{2ik_zh}\right]E_1(k_\rho)\mathrm{d}k_\rho$$

From this, we obtain:

$$E_1(k_\rho)e^{2ik_zh} + \frac{ik_\rho^3\alpha}{4\pi k_z}\left[1 + e^{2ik_zh}\right]\int_0^{k_0}\left[1 + e^{2ik'_zh}\right]E_1(k'_\rho)\mathrm{d}k'_\rho = 0$$

which is **Equation 3** of the paper. Equivalently, we may write:

$$p = -\frac{i\alpha}{4\pi}\int_0^{k_0}\frac{(1 + e^{2ik_zh})^2 p}{e^{2ik_zh}}\frac{k_\rho^3}{k_z}\mathrm{d}k_\rho$$

Dividing both sides by p gives **Equation 4** of the paper.